

## H Question Salience Annotation Guidelines

**Motivation.** When summarizing long texts, we must consciously decide what information to include or exclude from a summary. These decisions are grounded in a notion of information salience, or how important we consider the information for our intended audience. We study this phenomenon in the context of automatic text summarization systems. Specifically, we aim to understand how well these systems replicate the judgments of domain experts regarding what information is most relevant.

**Task.** Imagine you are asked to **summarize a paper describing the results of a randomized controlled trial (RCT)** for a typical reader in this field. The summary should provide enough context to stand alone, since the reader will only see your summary and no other parts of the paper. Furthermore, the summary length is constrained, requiring you to think about what content to prioritize. In this study, we frame content as questions that a summary could answer.

Ask yourself: **What are some key questions you want the summary to answer?** Your task is to rate the relative importance of a list of questions on the following scale.

- ☐ (1) Least important; I would exclude this information from a summary.
- ☐ (2) Low importance; I would include this information if there is room.
- ☐ (3) Medium importance; I would probably include this information.
- ☐ (4) High importance; I would definitely include this information.
- ☐ (5) Most important; One of the first questions to be answered in the summary.

**Rationale.** For each rating, please provide a brief (1-sentence) rationale explaining your decision or highlighting any considerations or uncertainties.

**Example answers.** To give you a feeling for the kind of content a question might elicit, all questions have an illustrative answer sourced from a randomly chosen document (= RCT paper). Please keep the following in mind:

- *Answer length* does not determine the question's importance.
- *Phrasing and selection.* The precise answer phrasing can be different in the summary, and

not all answer content must appear in the summary.

- *Overlap.* Some questions may elicit overlapping answers. Therefore, focus on the essence of each question. Remember that in an actual summary, overlapping answer information would only be stated once, so don't worry about it (see below).
- *Relevance.* The questions are answerable with most documents in this genre. Do your rating on the assumption that the document talks about this information.

### Suggested process.

1. Read all questions first.
2. Identify questions that seem most/least important, and rate these as "anchor points."
3. Then, rate the remaining questions.

Finally, there are no right or wrong ratings. Use your best judgment and intuition. Thank you for participating!

### Appendix: Example of overlapping answers.

Consider questions Q1–Q3 below. Each question asks for a distinct unit of information, but the answer of Q3 overlaps with the answer of Q1 and Q2. The overlapping information is highlighted in orange while the *essence of the question* is highlighted in green. Base your rating on the essence of the question.

#### Example of overlapping answers

**Q1. What was the study design or setting of the trial?** This trial is a multicentre, randomized, double-blind, phase 3 study.

**Q2. What specific treatments were compared in the study?** DBPR108 100 mg, sitagliptin 100 mg, and placebo.

**Q3. How were the participants or subjects of the study selected and divided?** In this multicentre, randomized, double-blind, phase 3 study, adult patients with type 2 diabetes were randomly assigned to receive either DBPR108 100mg, sitagliptin 100mg, or placebo once daily. A total of 766 patients were enrolled and divided into three groups: DBPR108 100mg (n=462), sitagliptin 100mg (n=152), or placebo (n=152).

# Question Salience in Text Summarization

**Task.** Imagine you are asked to **summarize the discussion section of an astro-physics paper** for a typical reader in this field. The summary should provide enough context to stand alone, since the reader will *only* see your summary and no other parts of the paper. What are some key questions you want the summary to answer? Here, your task is to rate the (relative) importance of a list of questions that could be answered in the summary.

**Rating scale.**

- 1. Least important; I would exclude this information from a summary.
- 2. Low importance; I would include this information if there is room.
- 3. Medium importance; I would probably include this information.
- 4. High importance; I would definitely include this information.
- 5. Most important; One of the first questions to be answered in the summary.

**Duration.** Please keep track of how long it took you to do the rating.

## Questions

Show all examples

**What is the main focus of the study?**

The main focus of the study is to test cosmic evolution of SNe Ia, specifically to quantify systematics from any evolution of intrinsic properties with the age of the universe, which is crucial for precision probes of dark energy.

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What detailed evidence or data is used to support the study's claims?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What broader context or field does the study contribute to or address?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**Which future research directions does the study recommend or outline?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**How do the findings affect the understanding of astronomical systems?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What comparisons are made within the study?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What methodology or techniques are employed in the study?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What specific challenges or limitations does the study address or identify?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What are the broader implications or potential applications of the findings?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What specific phenomena or processes are being investigated in the study?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**How do the findings relate to existing models or theories?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What are the main findings of the study?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

**What specific variables or conditions are crucial in the study's findings?**

○ ○ ○ ○ ○  
1 2 3 4 5

Rationale

Any comments (optional)...

Figure 17: Interface for question salience annotation. Each question can be expanded to show an illustrative answer sourced from a randomly chosen document. The questions shown here are for the *Astro* dataset.



Figure 18: Fully worked example of the question-based content analysis. Two documents in a fictional domain are each summarized at three lengths. Afterwards Steps 1 – 4 are analogous to § 2.2. Summary claims are color-coded.